

FA 1: Examining Data, Transformation, and Regression

2026-08-30

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2     4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr       1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(gapminder)
```

```
# Select the 2007 observations
data_2007 <- gapminder %>%
  filter(year == 2007)
```

```
# View the data
head(data_2007)
```

```
## # A tibble: 6 x 6
##   country    continent  year lifeExp    pop gdpPercap
##   <fct>      <fct>    <int> <dbl>    <int>    <dbl>
## 1 Afghanistan Asia      2007  43.8 31889923    975.
## 2 Albania    Europe    2007  76.4  3600523   5937.
## 3 Algeria    Africa    2007  72.3 33333216   6223.
## 4 Angola     Africa    2007  42.7 12420476   4797.
## 5 Argentina  Americas  2007  75.3 40301927  12779.
## 6 Australia  Oceania   2007  81.2 20434176  34435.
```

3.1. Part A – Initial Examination of the Data

A(a)

```
# Number of observations
nrow(data_2007)
```

```
## [1] 142
```

```
# Number of variables
ncol(data_2007)
```

```
## [1] 6
```

```
# Structure of the dataset
str(data_2007)
```

```
## tibble [142 x 6] (S3: tbl_df/tbl/data.frame)
## $ country : Factor w/ 142 levels "Afghanistan",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ continent: Factor w/ 5 levels "Africa","Americas",...: 3 4 1 1 2 5 4 3 3 4 ...
## $ year      : int [1:142] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
## $ lifeExp   : num [1:142] 43.8 76.4 72.3 42.7 75.3 ...
## $ pop       : int [1:142] 31889923 3600523 33333216 12420476 40301927 20434176 8199783 708573 150448
## $ gdpPercap: num [1:142] 975 5937 6223 4797 12779 ...
```

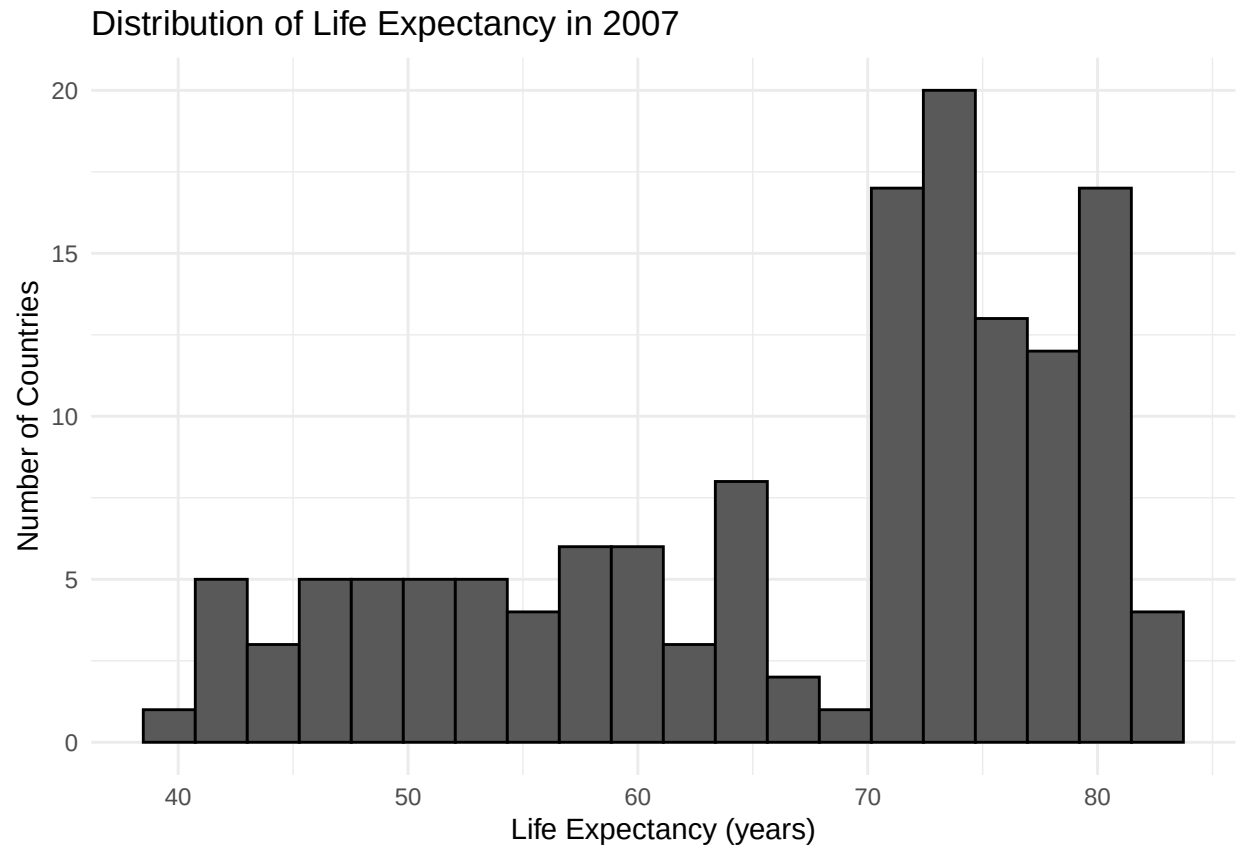
A(b)

```
# Summary statistics
summary(data_2007 %>%
  select(lifeExp, gdpPercap))
```

```
##      lifeExp      gdpPercap
## Min.   :39.61   Min.    : 277.6
## 1st Qu.:57.16   1st Qu.: 1624.8
## Median :71.94   Median : 6124.4
## Mean   :67.01   Mean    :11680.1
## 3rd Qu.:76.41   3rd Qu.:18008.8
## Max.   :82.60   Max.    :49357.2
```

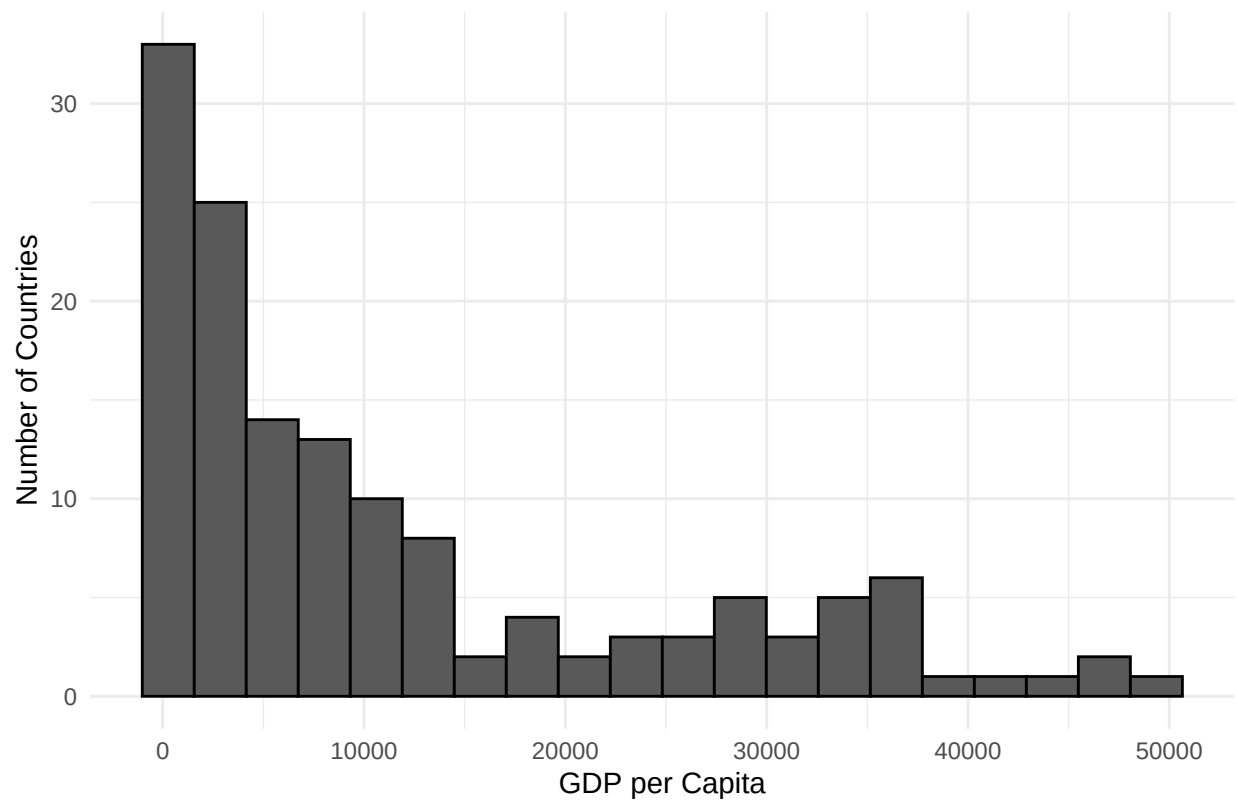
A(c)

```
ggplot(data_2007, aes(x = lifeExp)) +
  geom_histogram(bins = 20, color = "black") +
  labs(
    title = "Distribution of Life Expectancy in 2007",
    x = "Life Expectancy (years)",
    y = "Number of Countries"
  ) +
  theme_minimal()
```



```
ggplot(data_2007, aes(x = gdpPercap)) +  
  geom_histogram(bins = 20, color = "black") +  
  labs(  
    title = "Distribution of GDP per Capita in 2007",  
    x = "GDP per Capita",  
    y = "Number of Countries"  
  ) +  
  theme_minimal()
```

Distribution of GDP per Capita in 2007

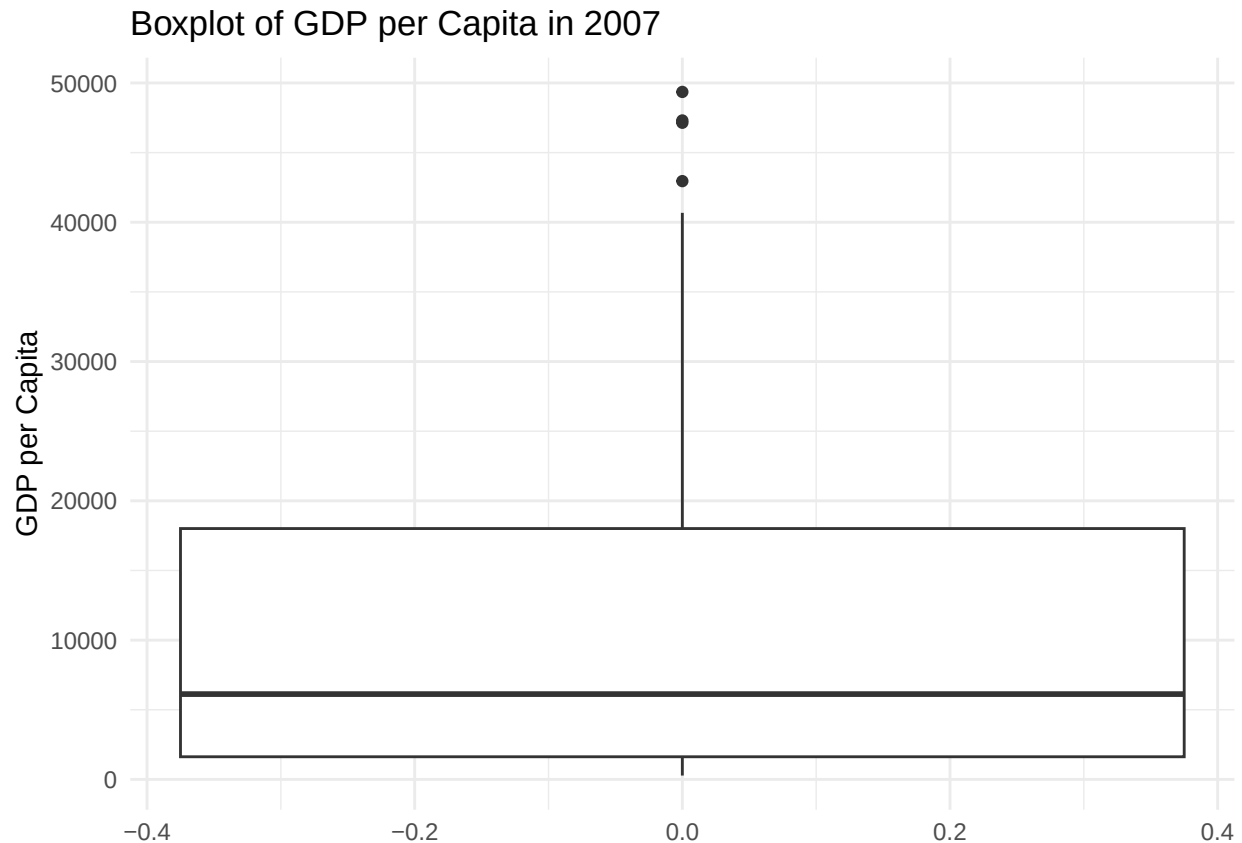


$A(d)$

```
ggplot(data_2007, aes(y = lifeExp)) +
  geom_boxplot() +
  labs(
    title = "Boxplot of Life Expectancy in 2007",
    y = "Life Expectancy (years)"
  ) +
  theme_minimal()
```



```
ggplot(data_2007, aes(y = gdpPercap)) +  
  geom_boxplot() +  
  labs(  
    title = "Boxplot of GDP per Capita in 2007",  
    y = "GDP per Capita"  
  ) +  
  theme_minimal()
```



3.2. Part B – Examine the Bivariate Relationship

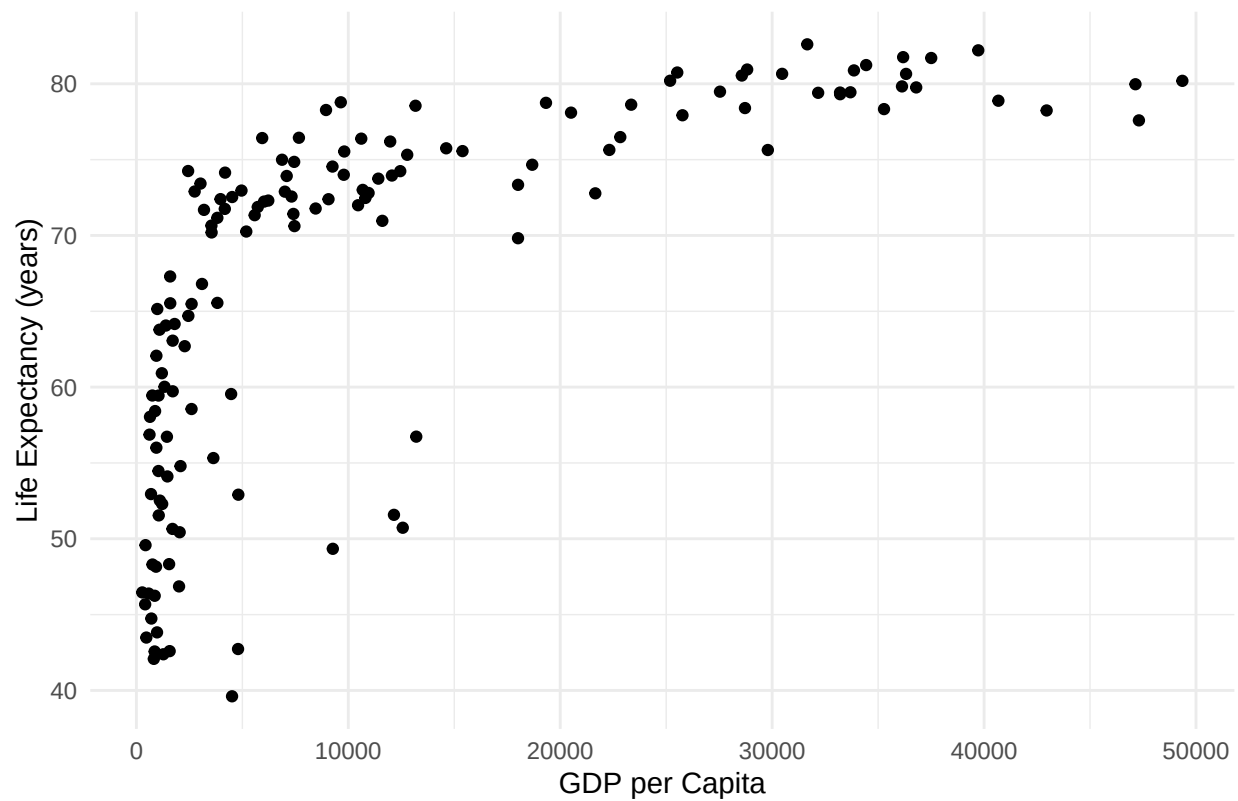
B(a)

The scatterplot shows a positive relationship between GDP per capita and life expectancy. The relationship appears nonlinear because life expectancy increases quickly at lower GDP levels and gradually levels off at higher GDP levels. The relationship is moderately to strongly associated, with more spread among countries with lower GDP per capita.

B(b)

```
ggplot(data_2007, aes(x = gdpPercap, y = lifeExp)) +
  geom_point() +
  labs(
    title = "Life Expectancy vs GDP per Capita in 2007",
    x = "GDP per Capita",
    y = "Life Expectancy (years)"
  ) +
  theme_minimal()
```

Life Expectancy vs GDP per Capita in 2007

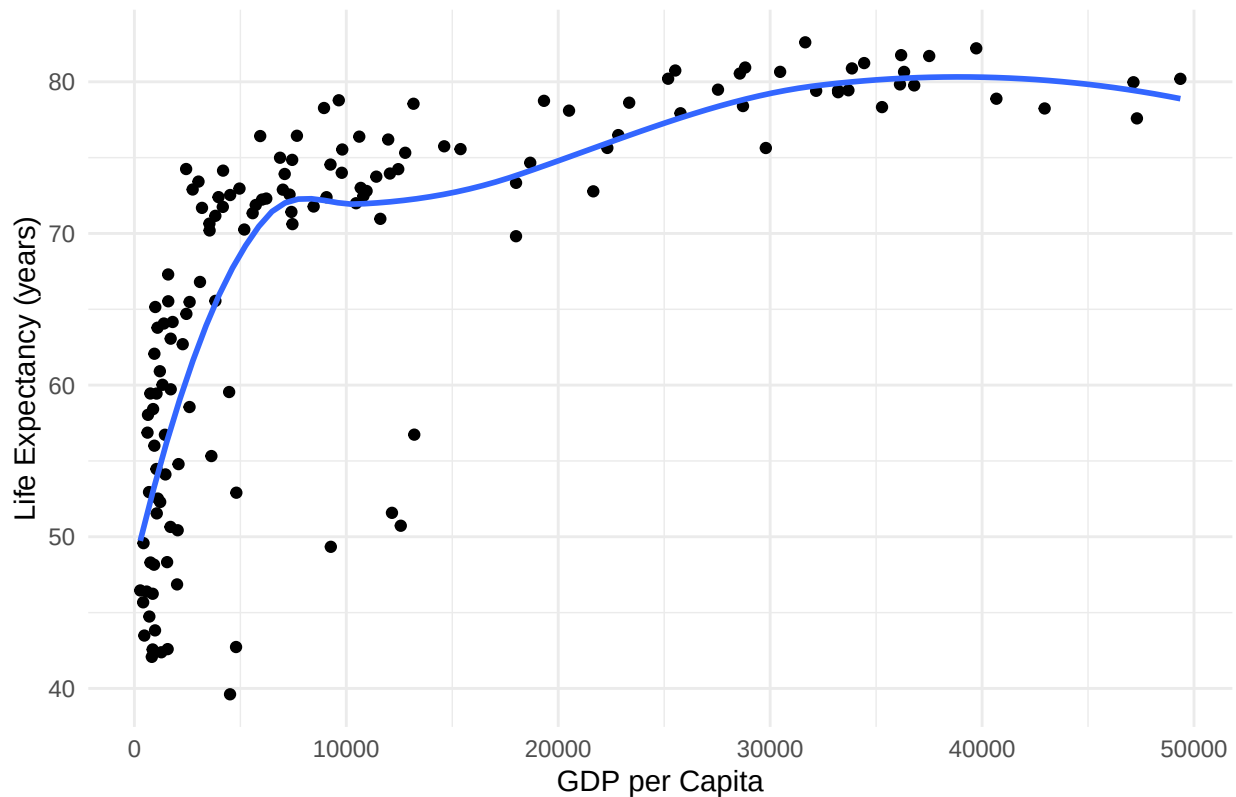


B(c)

```
ggplot(data_2007, aes(x = gdpPercap, y = lifeExp)) +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE) +
  labs(
    title = "Life Expectancy vs GDP per Capita with LOESS",
    x = "GDP per Capita",
    y = "Life Expectancy (years)"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Life Expectancy vs GDP per Capita with LOESS



B(d)

It is important to examine the scatterplot before fitting a regression model because graphs can reveal patterns that numerical summaries may not show. The scatterplot helps us identify nonlinearity, unequal spread, clusters, and unusual observations. This relates to Anscombe's Quartet, which shows that data can have similar numerical summaries but different graphical patterns.

3.3. Part C – Transformation of the Explanatory Variable

C(a)

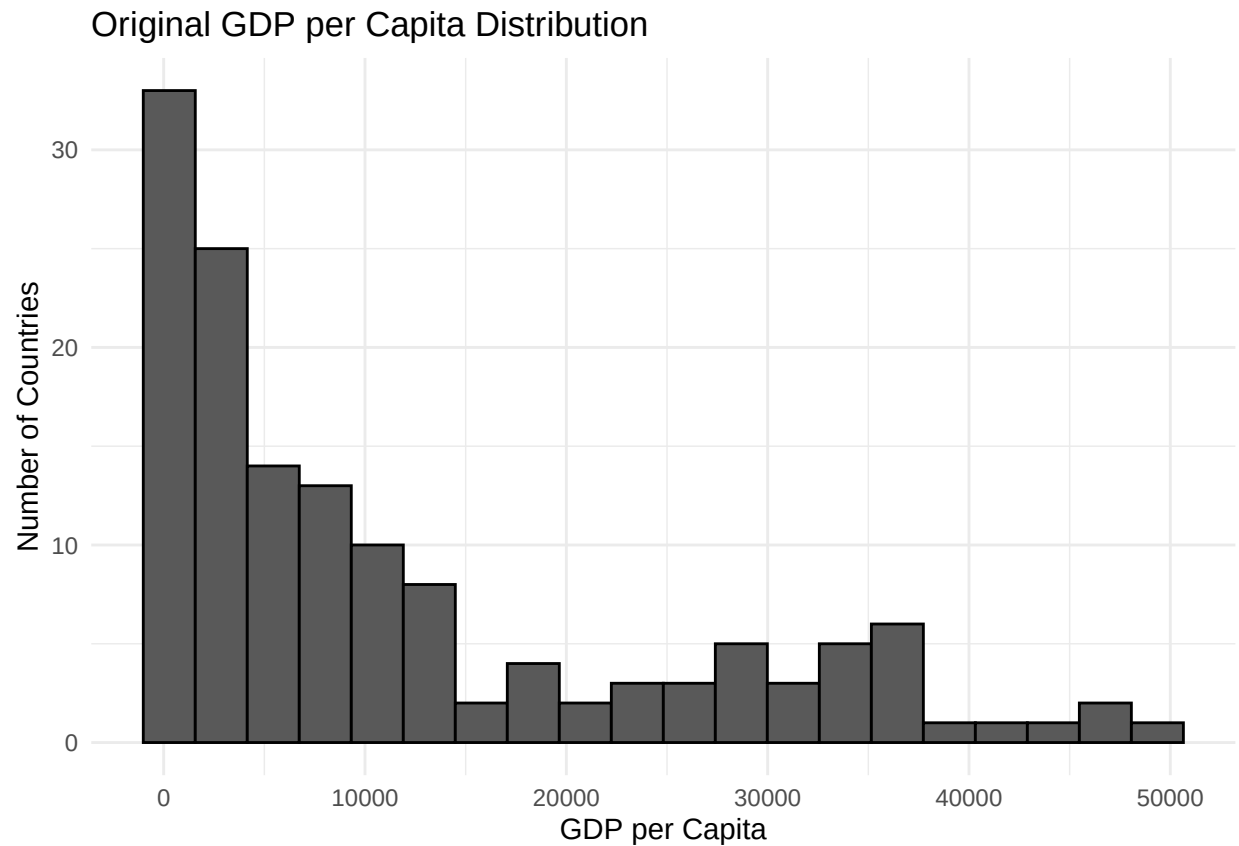
```
data_2007 <- data_2007 %>%
  mutate(log_gdpPercap = log(gdpPercap))

head(data_2007 %>%
  select(gdpPercap, log_gdpPercap))
```

```
## # A tibble: 6 x 2
##   gdpPercap log_gdpPercap
##   <dbl>      <dbl>
## 1    975.         6.88
## 2   5937.         8.69
## 3   6223.         8.74
## 4   4797.         8.48
## 5  12779.         9.46
## 6 34435.        10.4
```

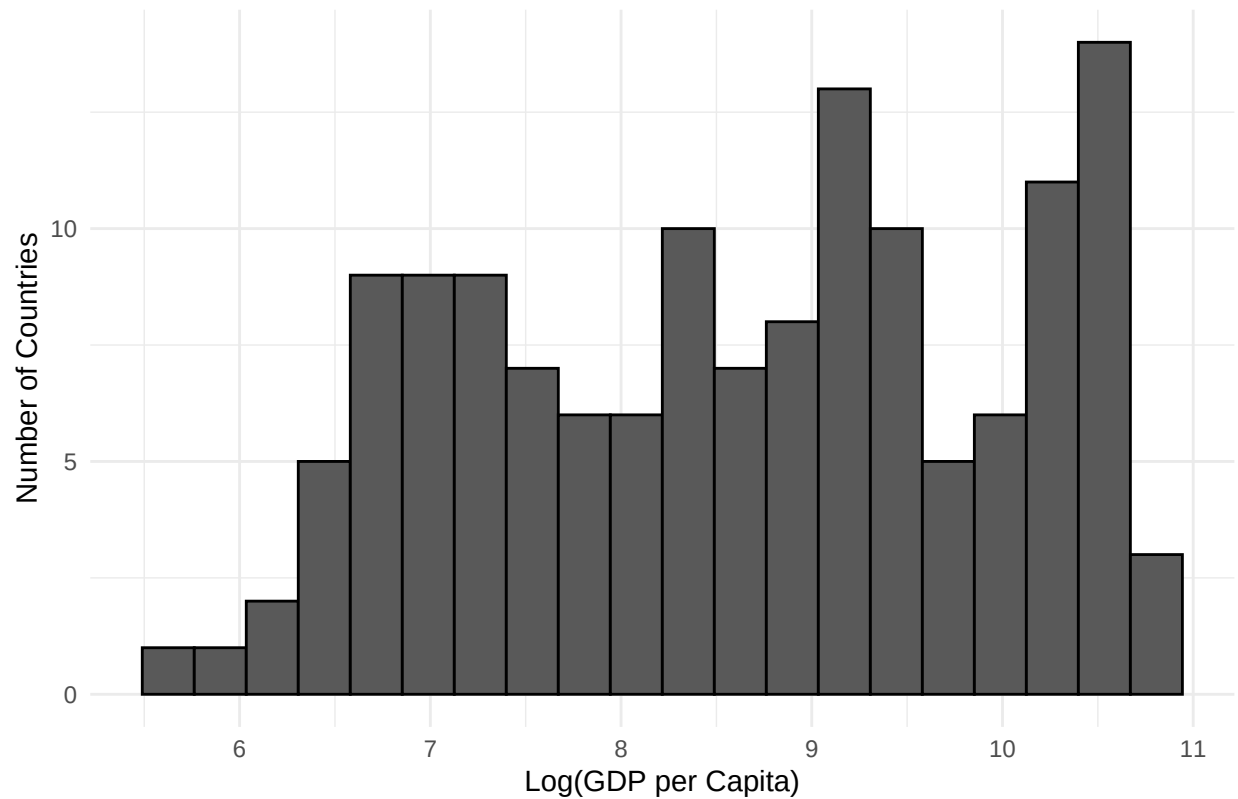
C(b)


```
ggplot(data_2007, aes(x = gdpPerCap)) +
  geom_histogram(bins = 20, color = "black") +
  labs(
    title = "Original GDP per Capita Distribution",
    x = "GDP per Capita",
    y = "Number of Countries"
  ) +
  theme_minimal()
```



```
ggplot(data_2007, aes(x = log_gdpPerCap)) +
  geom_histogram(bins = 20, color = "black") +
  labs(
    title = "Log-Transformed GDP per Capita Distribution",
    x = "Log(GDP per Capita)",
    y = "Number of Countries"
  ) +
  theme_minimal()
```

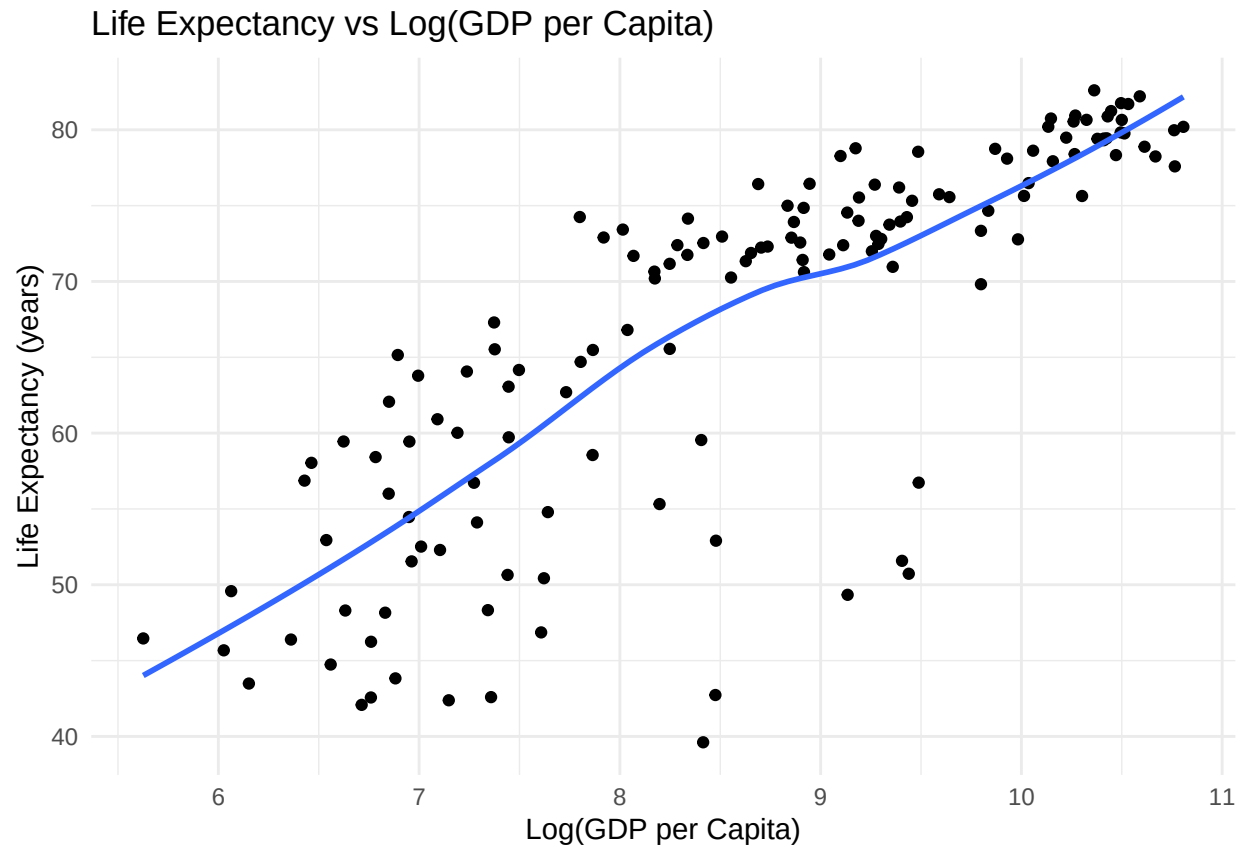
Log-Transformed GDP per Capita Distribution



C(c)

```
ggplot(data_2007, aes(x = log_gdpPercap, y = lifeExp)) +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE) +
  labs(
    title = "Life Expectancy vs Log(GDP per Capita)",
    x = "Log(GDP per Capita)",
    y = "Life Expectancy (years)"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



$C(d)$

```
# Original GDP
gdp_stats <- boxplot.stats(data_2007$gdpPercap)$stats

gdp_lower <- gdp_stats[2]
gdp_median <- gdp_stats[3]
gdp_upper <- gdp_stats[4]

gdp_ratio <- (gdp_upper - gdp_median) /
             (gdp_median - gdp_lower)

gdp_ratio

## [1] 2.625882

# Log GDP
log_stats <- boxplot.stats(data_2007$log_gdpPercap)$stats

log_lower <- log_stats[2]
log_median <- log_stats[3]
log_upper <- log_stats[4]

log_ratio <- (log_upper - log_median) /
             (log_median - log_lower)
```

```
log_ratio
```

```
## [1] 0.803147
```

3.4. Part D – Compare Regression Descriptions

D(a)

```
model1 <- lm(lifeExp ~ gdpPercap, data = data_2007)
```

```
summary(model1)
```

```
##
## Call:
## lm(formula = lifeExp ~ gdpPercap, data = data_2007)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.828  -6.316   1.922   6.898  13.128
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.957e+01  1.010e+00   58.95  <2e-16 ***
## gdpPercap    6.371e-04  5.827e-05   10.93  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.899 on 140 degrees of freedom
## Multiple R-squared:  0.4606, Adjusted R-squared:  0.4567
## F-statistic: 119.5 on 1 and 140 DF, p-value: < 2.2e-16
```

```
model2 <- lm(lifeExp ~ log_gdpPercap, data = data_2007)
```

```
summary(model2)
```

```
##
## Call:
## lm(formula = lifeExp ~ log_gdpPercap, data = data_2007)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25.947  -2.661   1.215   4.469  13.115
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.9496    3.8577   1.283   0.202
## log_gdpPercap  7.2028    0.4423  16.283  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.122 on 140 degrees of freedom
## Multiple R-squared:  0.6544, Adjusted R-squared:  0.652
## F-statistic: 265.2 on 1 and 140 DF, p-value: < 2.2e-16
```

$D(b)$

```
# Model 1  
summary(model1)$r.squared
```

```
## [1] 0.4605827
```

```
summary(model1)$sigma
```

```
## [1] 8.898646
```

```
# Model 2  
summary(model2)$r.squared
```

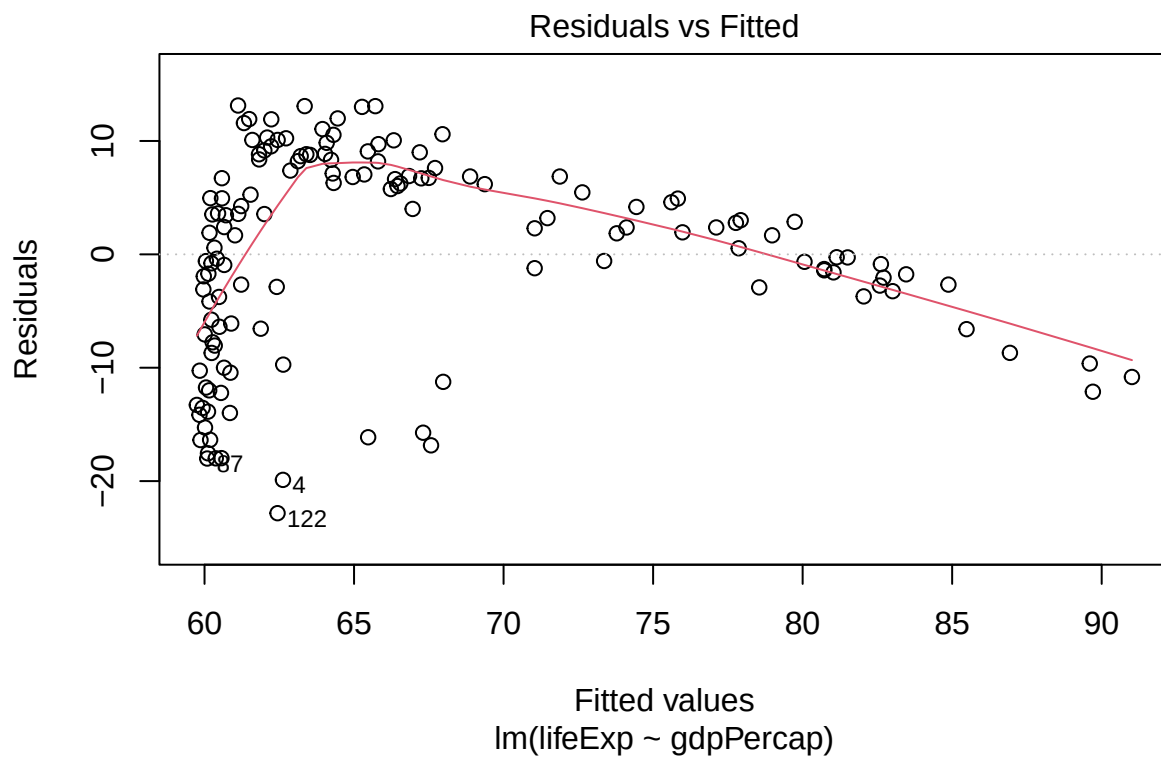
```
## [1] 0.654449
```

```
summary(model2)$sigma
```

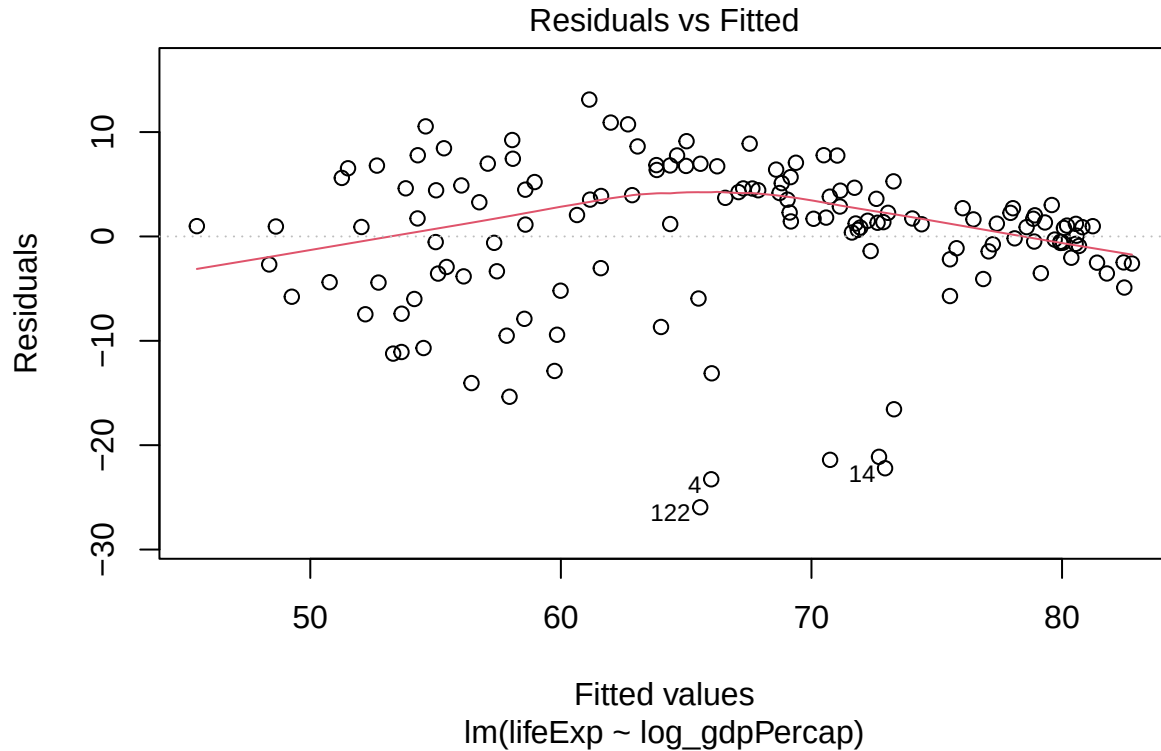
```
## [1] 7.122255
```

$D(c)$

```
plot(model1, which = 1)
```



```
plot(model2, which = 1)
```



D(d)

Model 2, which uses $\log(\text{GDP per capita})$, is considered more appropriate for describing the relationship between life expectancy and GDP per capita. The log-transformed model provides a more suitable description because the relationship becomes more nearly linear and the spread of the residuals is more consistent. In addition, the numerical measures such as R^2 and residual standard error can be compared to determine whether the transformed model provides a better fit. Therefore, the log transformation improves the suitability of the regression model.

D(e)

One important limitation is that the regression analysis shows an association between GDP per capita and life expectancy, but it does not prove that higher GDP per capita causes higher life expectancy. Other factors, such as healthcare, education, nutrition, and living conditions, may also affect life expectancy.

Conclusion

The analysis showed a positive relationship between GDP per capita and life expectancy among countries in 2007. Countries with higher GDP per capita generally tended to have higher life expectancy, although the relationship was not completely linear on the original GDP scale. The GDP per capita variable was also strongly right-skewed, so a logarithmic transformation was examined. The log transformation made the distribution more symmetric and provided a more nearly linear relationship between GDP per capita and life expectancy. Based on the graphical and numerical comparisons, the regression model using $\log(\text{GDP per capita})$ provides a more appropriate description of the relationship. However, this analysis only shows an association and does not establish that higher GDP per capita causes higher life expectancy, since other factors may also influence life expectancy.